

Lierda NT26-KCN F Series (KR) Hardware Design Manual

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Revision History

Document Version	Change date	Reviser	Change content
Rev1.0	24-12-26	Cheney Chen	Initial Version
Rev1.1	25-07-01	Cheney Chen	Modify the shutdown precautions; Modify the ADC sampling range; Modify that the serial port RX does not have reverse voltage protection function; Add a section "1.1 Special Symbols" Modify the power consumption description of the sleep mode. Change the reference ground of the third design in Section 5.4, Table 5-5 to L4 layer.
Rev1.2	25-08-22	Cheney Chen	Partial image updates in English.
Rev1.3	25-09-26	Cheney Chen	Modify the USIM section. Change the USIM_VDD pin number to 14.
Rev1.4	25-10-21	Cheney Chen	The 0R resistor in the antenna reference circuit diagram has been replaced by a 100pF capacitor. Update the reflow soldering operation instruction diagram.

Safety Instructions

Users are responsible for complying with relevant regulations on wireless communication modules and devices in other countries and specific environmental regulations for use. By following the following safety principles, personal safety can be ensured and help protect products and work environments from potential damage. Our company is not responsible for any losses resulting from customers' failure to comply with these regulations.



Safety first on the road! Do not use handheld mobile devices while driving unless they have a hands-free function. Please pull over before making a call!



Please turn off your mobile devices before boarding. The wireless function of mobile devices is prohibited from being turned on in the aircraft to prevent interference with the aircraft communication system. Ignoring this prompt may compromise flight safety and even violate the law.



When in a hospital or healthcare facility, pay attention to any restrictions on the use of mobile terminal devices. RF interference can cause medical equipment to malfunction, so it may be necessary to turn off mobile terminal devices.



Mobile terminal devices do not guarantee effective connection in all situations, such as when the mobile terminal device has no credit or the SIM card is invalid. When you encounter the above situations in an emergency, remember to use emergency calls, and ensure that your device is powered on and in an area with sufficient signal strength.



Your mobile terminal device will receive and emit radio frequency signals when it is powered on, which may cause radio frequency interference when it is close to a TV, radio, computer, or other electronic devices.



Please keep mobile terminal devices away from flammable gases. When you are near gas stations, oil depots, chemical plants, or explosive operation sites, please turn off the mobile terminal devices. Operating electronic devices in any potentially explosive hazardous area poses a safety risk.

Module selection for application

Serial number	Module model	Product model	Support frequency band	Dimensions (mm)	Module introduction
1	NT26-KCN	NT26KCNF20NNK	Band1/3/5/8	17.7×15.8×2.4	Support low power consumption serial port. Sleep mode: VDD_EXT power off

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1 Introduction

This document defines the application specifications of the Lierda NT26-KCN F series Cat.1 wireless communication module, describing its frequency band functions, key features, hardware interfaces, operating modes, electrical characteristics, mechanical specifications, packaging, storage, etc., which can help users quickly grasp the application method of this Cat.1 module and efficiently and flexibly carry out product design.

1.1 Special characters

Table 1-1 Special Symbol Description

Symbol	Definition
※	Unless otherwise specified, the asterisk (*) marked after the module's functions, features, interfaces, pin names, AT commands, or parameters indicates that the development of the module's function, feature, interface, pin name, AT command, or parameter is in progress and not supported at the moment.

2 Product Overview

The NT26-KCN F series module is a leading Cat.1 wireless communication module globally, compliant with 3GPP R14 Cat.1bis standard, supporting bandwidths of 1.4/3/5/10/15/20MHz, characterized by small size, low power consumption, strong anti-interference capability, etc. The module is suitable for various common IoT application scenarios, such as:

- ◆ Smart meter reading
- ◆ Intelligent Parking
- ◆ Smart City
- ◆ Intelligent Security
- ◆ Asset Tracking
- ◆ Smart Home Appliances
- ◆ Agriculture and environmental monitoring, etc.

2.1 Frequency bands and functions

The wireless network functions supported by the NT26-KCN F series module are as follows:

Table 2-1 Module Supports Wireless Network Functions

Network Function	Explanation
LTE-FDD	B1/B3/B5/B8

2.2 Key Features

Key features of the NT26-KCN F series module are as follows:

Table 2-2 Module Key Features Description

Parameters	Explanation
Module packaging	LCC+LGA
Module size	17.7mm×15.8mm×2.4mm (L×W×H)
Module weight	About 1.3 grams
Operating voltage	◆ Supply Voltage: 3.3V to 4.5V, Typical Value: 3.8V ◆ Extended power supply voltage(1): 3.1~4.5V
Operating temperature	◆ Normal operating temperature range(2): -35~75℃ ◆ Extended operating temperature range(3): -40 to 85℃
Storage temperature	Storage temperature range(4): -40~90℃
VDD_EXT characteristics	The F series offers two hardware versions. Sleep mode: VDD_EXT power off, driving capability 120mA Sleep mode: VDD_EXT does not power off, driving capability 3mA
Ultra-low power	Sleep mode: 2.5uA@3.8V
Transmit power	23dBm (Max)
Antenna interface	50 ohm characteristic impedance
USIM	◆ Supported USIM card types: Class B (3.0 V) and Class C (1.8 V) ◆ Supported USIM2 card types: Currently only supports Class C (1.8 V)
UART	Main Serial Port: ◆ Used for AT command transmission and data transfer, the supported baud rate defaults to 115200bps. ◆ Used for firmware upgrade, the supported baud rate defaults to 921600bps.
	Debug serial port: ◆ Used for software debugging debugging ◆ The default baud rate of the serial port is 3Mbps.
	Auxiliary serial port: ◆ User-defined function
USB	◆ Compatible with USB 2.0 (supporting slave mode only), with a maximum data transfer rate of up to 480 Mbps. ◆ Used for AT command transmission, data transfer, software debugging, and firmware upgrade

I2C	Support I2C function
SPK	Support simulation audio output
I2S	Support I2S function
Indicator light	◆ Running status indicator ◆ Network status indicator
Communication interface characteristics	Support 3GPP Rel.13/14 Cat.1 wireless communication interface and protocol.
Characteristics of network protocols	Support protocols such as TCP/UDP/HTTP(S)/SSL/MQTT(S)/FOTA/PPP/RNDIS/FTP/ECM.
Data transmission characteristics	◆ LTE-FDD: maximum downlink speed of 10Mbps, maximum uplink speed of 5Mbps
Firmware upgrade	◆ Main serial port upgrade ◆ USB interface upgrade ◆ DFOTA upgrade
RoHS	All components comply with EU RoHS standards.

Note

- (1) See section 6.2 for a specific description of the extended voltage.
- (2)(3)(4) See section 6.6 for specific descriptions regarding temperature.

2.3 Function Block Diagram

NT26-KCN F series module has rich peripheral interfaces and RF functions, the functional block diagram is as follows:

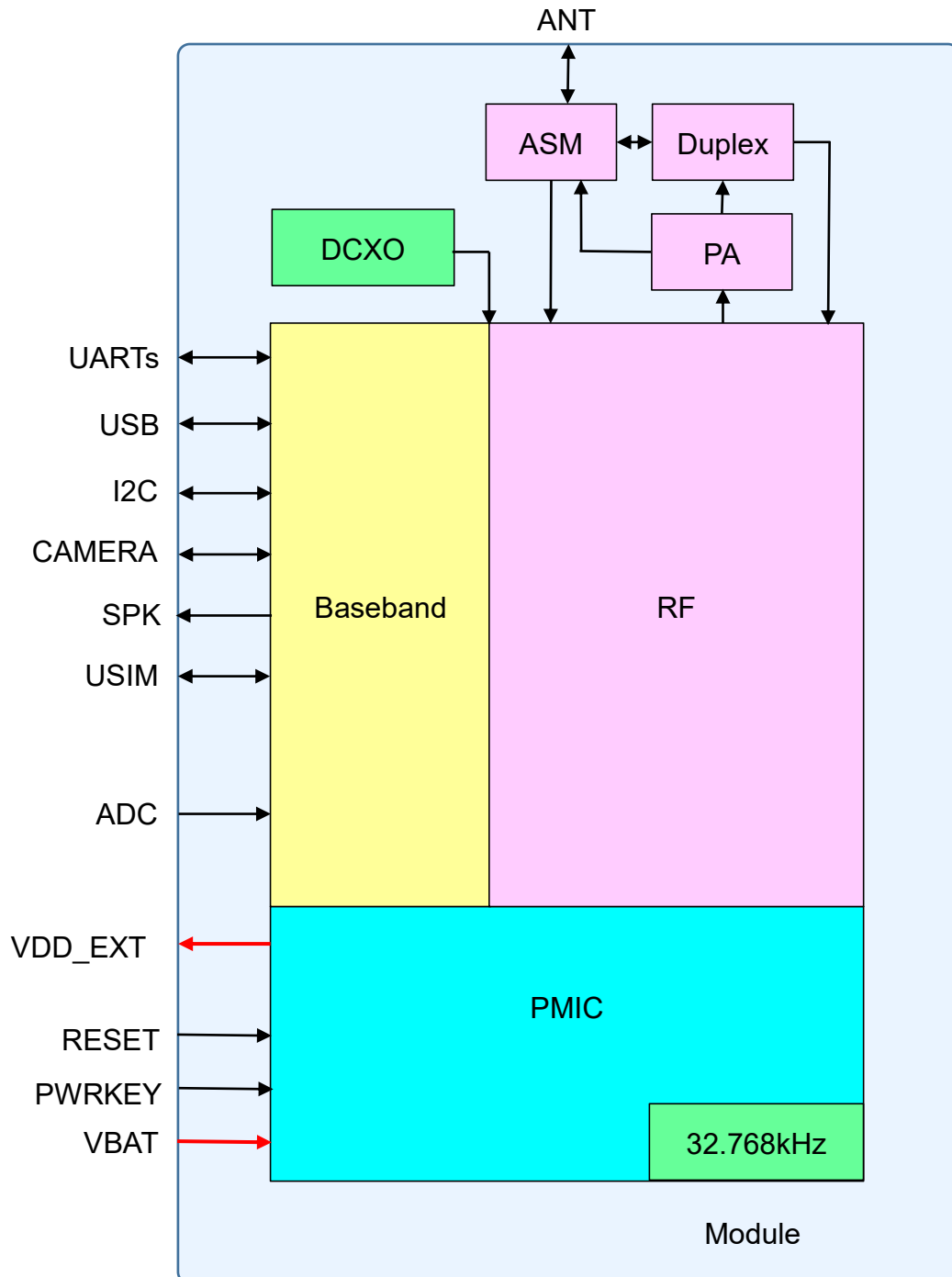


Figure 2-1 Module Function Block Diagram

2.4 Pinout diagram

The NT26-KCN F series module has a total of 109 pins, with 48 LCC pins and the remaining 61 LGA pins. The pin distribution diagram of the module is as follows:

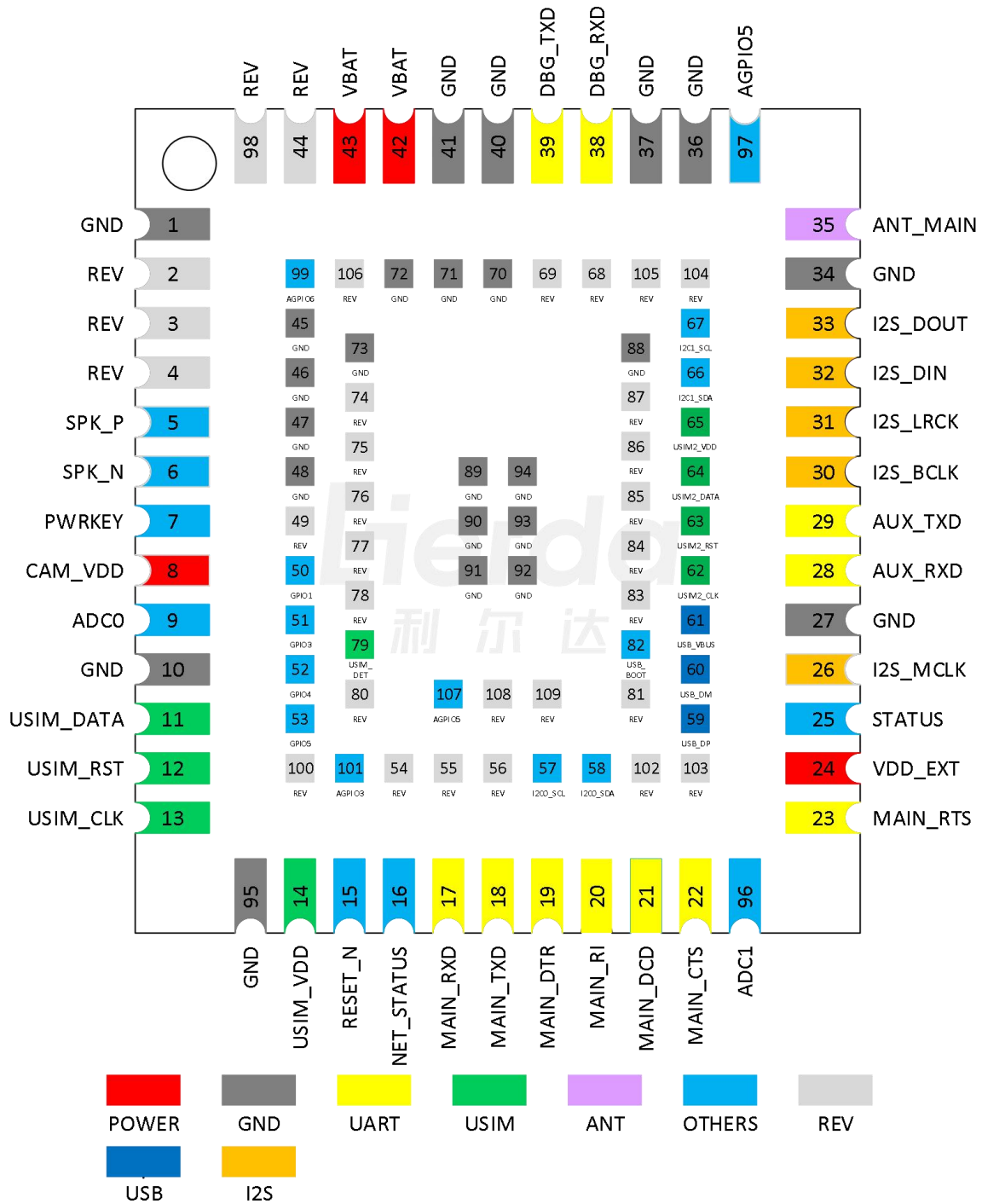


Figure 2-2 Module Pinout Diagram

2.5 Pin Description Table

To better understand the application, the following table provides a description of the type definitions for I/O parameters:

Table 2-3 Description of I/O Parameter Type Definitions

I/O parameter type	Explanation
DI	Numeric input
DO	Number output
DIO	Numeric input and output
AI	Simulated input
AO	Simulated output
AIO	Simulated input and output
PI	Power input
PO	Power output
G	Ground

Table 2-4 Module Power Domain Characteristics Description

Power domain parameters Type	DC characteristics	Explanation	Power supply object
VBAT	Vmax=4.5V Vmin=3.3V Vnorm=3.8V	Module power input, recommended to use 3.8V/1.2A power supply.	Module
VDD_EXT	VILmax=0.2×VDD_EXT VIHmin=0.7×VDD_EXT VOLmax=0.15×VDD_EXT VOHmin=0.8×VDD_EXT	Default 1.8V; Different hardware versions: In sleep mode, VDD_EXT power-off version, driving capability is 120mA; In sleep mode, the non-power-down version of VDD_EXT has a driving capability of 3mA.	VDD_EXT

VO_LDOIO	VOLmax=0.2×VO_LDOIO VOHmin=0.7×VO_LDOIO VILmax=0.15×VO_LDOIO VIHmin=0.8×VO_LDOIO	Default 1.8V, software configurable; Power off in sleep mode.	UART GPIO
VDD18AON	VILmax=0.2×VDD18AON VIHmin =0.7×VDD18AON VOLmax=0.15×VDD18AON VOHmin=0.8×VDD18AON	Not losing power in sleep mode.	WAKEUP RESET
LDO_AONIO	VILmax=0.2×LDO_AONIO VIHmin =0.7×LDO_AONIO VOLmax=0.15×LDO_AONIO VOHmin=0.8×LDO_AONIO	Default 1.8V, configurable by software; No power consumption in sleep mode.	AGPIO
VO_LDOSIM	Vnorm=1.8/3.0V VOLmax=0.15×VO_LDOSIM VOHmin=0.8×VO_LDOSIM VILmax=0.2×VO_LDOSIM VIHmin=0.7×VO_LDOSIM	SIM card dedicated power supply, supporting cards with 1.8/3.0V.	USIM

For a better understanding of the application, the table below explains the characteristics of the power domain parameters:

Table 2-5 Module Pin Description

1. Power					
Pin number	Pin names	I/O	Descri ption	DC characteristics	Note
42,43	VBAT	PI	Power supply	Vnorm=3.8V	External power supply needs to provide a current carrying capacity of 1.2A or above.
24	VDD_EXT	PO	Power output	Vnorm=1.8V Iomax=120mA/3mA	Can be used for external circuit pull-up or reference level; there are two modes of power off and non-power off in sleep mode; if not used, leave floating, it is

					recommended to reserve test points.
1,10,27,34, 36~37,40~41,45 ~48,70~72,73,88 ~95	GND	G	Ground	-	-

2. Power on/off and reset.

Pin number	Pin names	I/O	Describe	DC characteristics	Note
7	PWRKEY	DI	Module power on/off pins	$V_{ILmax}=0.45V$	Internal pull-up, active low, leave floating if not used.
15	RESET_N	DI	Module reset pin	$V_{ILmax}=0.3V$	Internal pull-up, low level effective, if not used, leave floating

3、USB

Pin number	Pin names	I/O	Describe	DC characteristics	Note
59	USB_DP	DIO	USB differential data (+)	-	If not used, leave floating
60	USB_DM	DIO	USB differential data (-)	-	
61	USB_VBUS	DI	USB_VBUS input wake-up	VDD18AON	Insert a module that can wake up the sleep state, if not used, leave floating.

4. Serial Port UART					
Pin number	Pin names	I/O	Description	DC characteristics	Note
17	MAIN_RXD	DI	Main serial port receives data	VO_LDOIO	If not used, leave floating
18	MAIN_TXD	DO	Main serial port sends data	VO_LDOIO	If not used, leave floating
19	MAIN_DTR	DI	Main serial port data terminal ready	VDD18AON	In sleep mode, the wake-up pin can be used.
20	MAIN_RI	DO	Main serial port outputs ringing prompt.	LDO_AONIO/ VDD18AON	AGPIO function, power domain LDO_AONIO, no power loss during sleep. WAKEUP function, power domain VDD18AON, no power loss during sleep.
21	MAIN_DCD	DO	Main serial port output carrier detection	VO_LDOIO	Cannot be used simultaneously with I2C1_SDA multiplexing.
22	MAIN_CTS	DO	Main serial port DTE clear to send	VO_LDOIO	Multiplexed with I2C0_SCL, cannot be used simultaneously
23	MAIN_RTS	DI	The main serial port DTE requests to send.	VO_LDOIO	Multiplexed with I2C0_SDA and I2S_MCLK, cannot be used simultaneously.

38	DBG_RXD	DI	Debugging serial port reception data	VO_LDOIO	If not used, leave floating
39	DBG_TXD	DO	Debugging serial port to send data	VO_LDOIO	If not used, leave floating
28	AUX_RXD*	DI	Assist in receiving data through the serial port.	VO_LDOIO	Multiplexed with I2S_DIN and GPIO4, cannot be used simultaneously
29	AUX_TXD*	DO	Assist in sending data through the serial port.	VO_LDOIO	Multiplexed with I2S_DOUT and GPIO5, cannot be used simultaneously.

5. External USIM card interface

Pin number	Pin names	I/O	Description	DC characteristics	Note
11	USIM_DATA	DIO	SIM card data cable	VO_LDOSIM	
12	USIM_RST	DIO	SIM card reset line	VO_LDOSIM	
13	USIM_CLK	DIO	SIM card clock line	VO_LDOSIM	
14	USIM_VDD	PO	SIM card power	VO_LDOSIM	1.8/3.0V
62	USIM2_CLK	DIO	SIM2 (sub card) clock line	VO_LDOIO	Not supported temporarily
63	USIM2_RST	DIO	SIM2 (secondary card) reset line	VO_LDOIO	Cannot be used simultaneously with GPIO1 multiplexing.
64	USIM2_DATA	DIO	SIM2 (secondary card) data cable	VO_LDOIO	Cannot be used simultaneously with GPIO3

					multiplexing.
65	USIM2_VDD	PO	SIM2 (Sub-card) power	VO_LDOSIM	Powered by the same source as USIM_VDD.
79	USIM_DET	DI	SIM card status signal detection	VDD18AON	High-level effective

6. Antenna interface RF

Pin number	Pin names	I/O	Description	DC characteristics	Note
35	ANT_MAIN	AIO	RF antenna interface	50Ω characteristic impedance	

I2S Interface

Pin number	Pin names	I/O	Description	DC characteristics	Note
26	I2S_MCLK	DO	I2S reference clock	VO_LDOIO	Multiplexed with I2C0_SDA and MAIN_RTS, cannot be used simultaneously
30	I2S_BCLK	DO	I2S clock output	LDO_AONIO	Cannot be used simultaneously with AGPIO3 multiplexing.
31	I2S_LRCK	DO	I2S synchronous signal	LDO_AONIO	If not used, leave floating
32	I2S_DIN	DI	I2S output input	VO_LDOIO	Multiplexed with AUX_RXD, GPIO4, cannot

					be used simultaneously.
33	I2S_DOUT	DO	I2S output output	VO_LDOIO	Multiplexed with AUX_TXD and GPIO5, cannot be used simultaneously

8. I2C interface

Pin number	Pin names	I/O	Description	DC characteristics	Note
57	I2C0_SDA	DIO	I2C serial data	VO_LDOIO	Cannot be used simultaneously with MAIN_RTS and I2S_MCLK.
58	I2C0_SCL	DO	I2C serial clock	VO_LDOIO	Cannot be used simultaneously with MAIN_CTS.
66	I2C1_SDA	DIO	I2C serial data	VO_LDOIO	Cannot be used simultaneously with MAIN_DCD.
67	I2C1_SCL	DO	I2C serial clock	VO_LDOIO	If not used, leave floating

9. SPK interface

Pin number	Pin names	I/O	Description	DC characteristics	Note
5	SPK_P/AU_OUT	DO	Audio differential output	VO_LDOIO	Cannot be used simultaneously with USB_BOOT

					multiplexing.
6	SPK_N	DO	Audio negative differential output	Internal ground	If not used, leave floating

10, ADC interface

Pin number	Pin names	I/O	Describe	DC characteristics	Note
9	ADC0	AI	General ADC interface	Internal direct connection Input voltage range: 0~1.05V	If not used, leave floating
96	ADC1	AI	Universal ADC interface	Internal partial pressure Input voltage range: 0~3.3V Default 1.05V Internal partial pressure can be activated through the software.	If not used, leave floating

10. Status Indicator

Pin number	Pin names	I/O	Describe	DC Characteristics	Note
16	NET_STATUS	DO	Network status indicator	LDO_AONIO	If not used, leave floating
25	STATUS	DO	Running status indicator	LDO_AONIO/ VDD18AON	AGPIO function, power domain LDO_AONIO, no power loss during sleep. WAKEUP function,

					power domain VDD18AON, no power loss during sleep
--	--	--	--	--	--

11. Other interfaces

Pin numb er	Pin names	I/O	Description	DC characteri stics	Note
50	GPIO1	DIO	General Purpose Input/Output (GPIO)	VO_LDOIO	Cannot be used simultaneously with USIM2_RST multiplexing.
51	GPIO3	DIO	General Purpose Input/Output (GPIO)	VO_LDOIO	Cannot be used simultaneously with USIM2_DATA multiplexing.
52	GPIO4	DIO	General Purpose Input/Output (GPIO)	VO_LDOIO	Cannot be used simultaneously with AUX_RXD and I2S_DIN.
53	GPIO5	DIO	General GPIO	VO_LDOIO	Multiplexed with AUX_TXD and I2S_DOUT, cannot be used simultaneously.
82	USB_BOOT	DI	Emergency download mode control. Pull up to VDD_EXT before the module is powered on, and the module will enter emergency download mode	VO_LDOIO	Multiplexed with SPK_P The module defaults to weak pull-down

			after startup.		internally and can function as a regular GPIO after boot-up.
97,107	AGPIO5	DIO	General-purpose GPIO, can be maintained in sleep mode	LDO_AONIO	If not used, leave floating
99	AGPIO6	DIO	General-purpose GPIO, can be maintained in sleep mode	LDO_AONIO	If not used, leave floating
101	AGPIO3	DIO	General-purpose GPIO, can be maintained in sleep mode.	LDO_AONIO	If not used, leave floating

12. Reserve interface

Pin number	Pin names	I/O	Description	DC characteristics	Note
2~4,44,49,54~56,68~69,74~78,80~81,83~87,98,100,102~106,108~109	REV	-	-	-	Undefined pins, recommend floating, not recommended for customer use (pins 68, 69, 75, 86, 87, 105 are prohibited from use, should be left floating).

2.6 Evaluation suite

Lierda can provide a complete evaluation and development kit, including module adapter boards, Cat.1 & NB-IoT module universal development boards, etc. Please feel free to contact us for consultation.

3 Working Characteristics

3.1 Working mode

The NT26-KCN F series module has multiple operating modes, and the following table briefly describes several common operating modes of the module:

Table 3-1 Working Modes

Pattern	Status Description
Normal working mode	Data Transmission: Network connection is normal. Module power consumption depends on network environment and data transmission rate.
	Idle status: The software is running normally. The module is connected to the network but not exchanging data with the network.
Minimum Function Mode	AT+CFUN=0 can set the module to minimum functionality mode, where both the RF and USIM card are not working.
Sleep mode	The power consumption of the module will be reduced to a minimum, but the module can still receive paging, short messages, and TCP/UDP data.
Shutdown mode	VBAT power supply is not disconnected, software stops working.

Note

Please refer to the AT command data sheet for the related AT+CFUN commands.

3.2 Sleep mode

Modules can enter sleep mode via AT commands. In sleep mode, the module can reduce power consumption to a very low level, while still being able to receive network information through the module's RF.

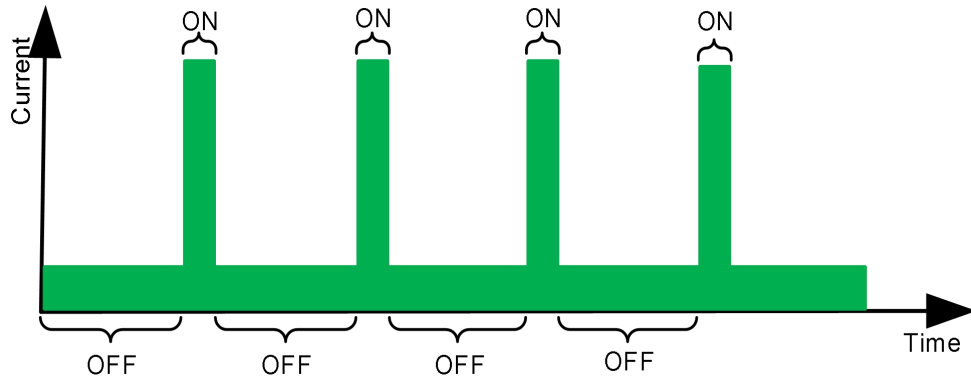


Figure 3-1 Power Consumption State of the Module in Sleep Mode

In sleep mode, the module peripherals are powered off to reduce power consumption, but the RF is in intermittent reception state and can still receive network information. When the module has a URC to report, it will wake up the host through the MAIN_RI pin.

3.2.1 Main serial port application scenarios

If the module and the host communicate via UART, two conditions must be met simultaneously to put the module into sleep mode:

- ◆ Enable sleep mode by executing the command AT+QSCLK=1;
- ◆ Ensure that MAIN_DTR remains at a high level or floating.

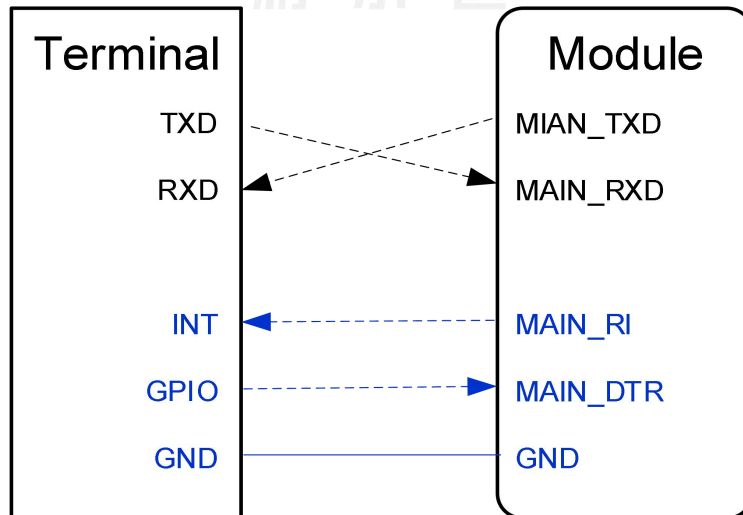


Figure 3-2 Application Circuit of Sleep Mode UART

When the module needs to report URC, the module notifies the host through MAIN_RI.

- ◆ The host pulling down MAIN_DTR can wake up the module.

If the host keeps pulling down MAIN_DTR, it can prevent the module from entering

sleep mode.

If the sleep function is enabled using AT+QSCLK=1, the module can be woken up by sending any command to the main UART interface.

3.3 Flight mode

When the module enters flight mode, the RF function is not available, and all AT commands related to RF cannot be accessed. The module can enter flight mode in the following ways:

Software method, you can set it by sending the AT+CFUN=<fun> command. The <fun> parameter can be set to 0 or 1.

- ◆ AT+CFUN=0: Minimum functionality mode (turn off RF function and USIM card).
- ◆ AT+CFUN=1: Full functionality mode (default).

3.4 Power supply design

3.4.1 Introduction to Power Supply

The NT26-KCN F series module provides 2 VBAT pins for connecting external power sources, and VDD_EXT is the power output of the module to the outside, which can be used as a reference voltage. The interface description is as follows:

Table 3-2 Power Supply Pin Definitions

Pin number	Pin names	Describe	Minimum value	Typical value	Maximum value	Unit
42,43	VBAT	Module power	3.3	3.8	4.5	V
24	VDD_EXT	Power output	-	1.8	-	V

The VBAT power supply range of the module is 3.3 to 4.5V, and it is necessary to ensure that the lowest input voltage is not lower than 3.3V when the module is working. The following diagram illustrates the situation of VBAT voltage drop during sudden transmission

of the module:

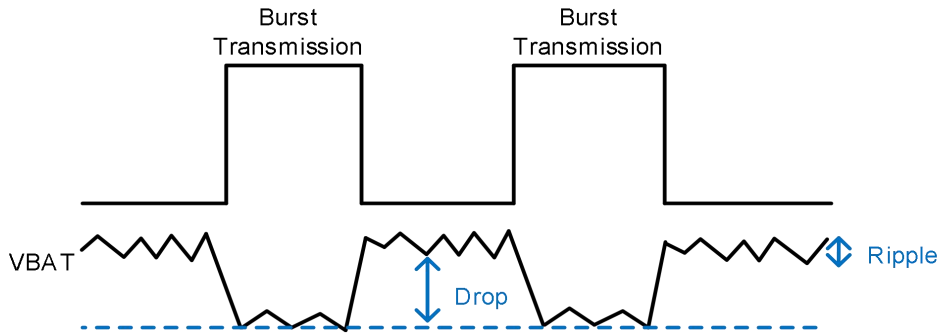


Figure 3-3 Emergency Transmission Power Requirements

Note

In sleep mode, VDD_EXT will power off.

3.4.2 Power supply design

When the module operates at the maximum transmission power in the 4G network, the transient operating current under the current network may reach 1.2A, which may cause a drop in the power supply voltage. In any case, it is necessary to ensure that the module's power supply voltage remains above 3.3V, otherwise the module may experience unexpected conditions such as rebooting. To reduce voltage drops, it is recommended to use a low ESR ($ESR < 0.7\Omega$) 100uF filtering capacitor. It is also recommended to reserve three chip multilayer ceramic capacitors (MLCC) with good ESR performance for VBAT, and the capacitors should be placed close to the VBAT pin. Additionally, it is recommended to add a high-power, low-clamping voltage, and high-reverse pulse current I_{pp} TVS diode at the VBAT terminal to improve the module's surge voltage withstand capability.

The power supply reference circuit is as follows:

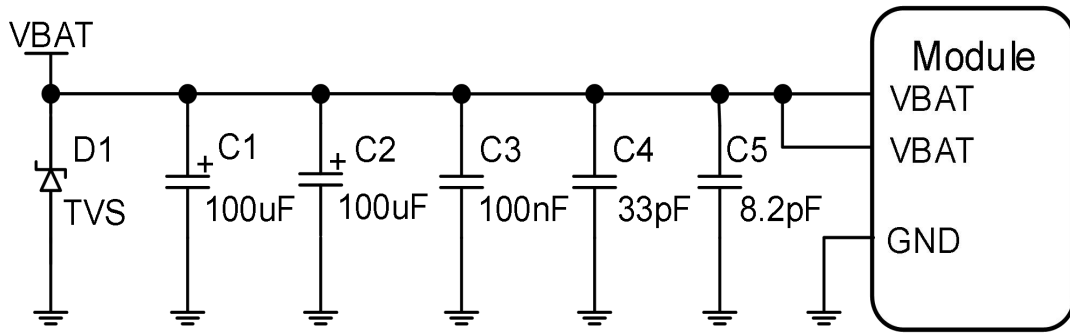


Figure 3-4 Module Power Supply Circuit Reference

In PCB design, the longer the VBAT trace, the wider it should be. It is recommended that the trace width should not be less than 1mm. The GND plane in the power section should be as complete as possible, with multiple GND vias. Additionally, decoupling capacitors should be placed as close as possible to the VBAT pin of the module.

- ◆ D1 is a TVS diode with high power and low clamping voltage;
- ◆ C1, C2 are 100uF low ESR tantalum capacitors, which enhance the power supply's ripple current capability and stabilize the voltage.
- ◆ C3~C5 are ceramic packaged filter capacitors with values of 100nF, 33pF, and 8.2pF respectively, used to eliminate high-frequency interference.

3.5 Power on.

3.5.1 Boot introduction

The module can be powered on through the PWRKEY pin, and the pin description is as follows:

Table 3-3 PWRKEY Pin Description

Pin number	Pin names	I/O	Description	Note
7	PWRKEY	DI	Module powered on, low level effective	Default to high level

The control logic of PWRKEY is as follows:

◆ The module can be powered on by pulling down the PWRKEY for at least 500ms in the shutdown state;

3.5.2 Power-on circuit reference design

It is recommended to use an open-drain drive circuit or directly control the power-on through a button. To prevent electrostatic discharge generated by contact, a TVS should be placed near the button for ESD protection. Please refer to the circuit below:

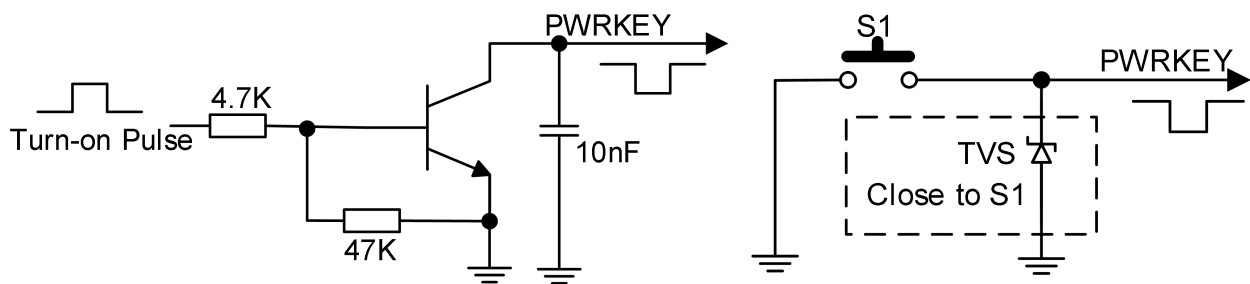


Figure 3-5 Power On/Off Circuit

3.5.3 Power-on sequence

The power-on sequence of the module is as shown in the figure below:

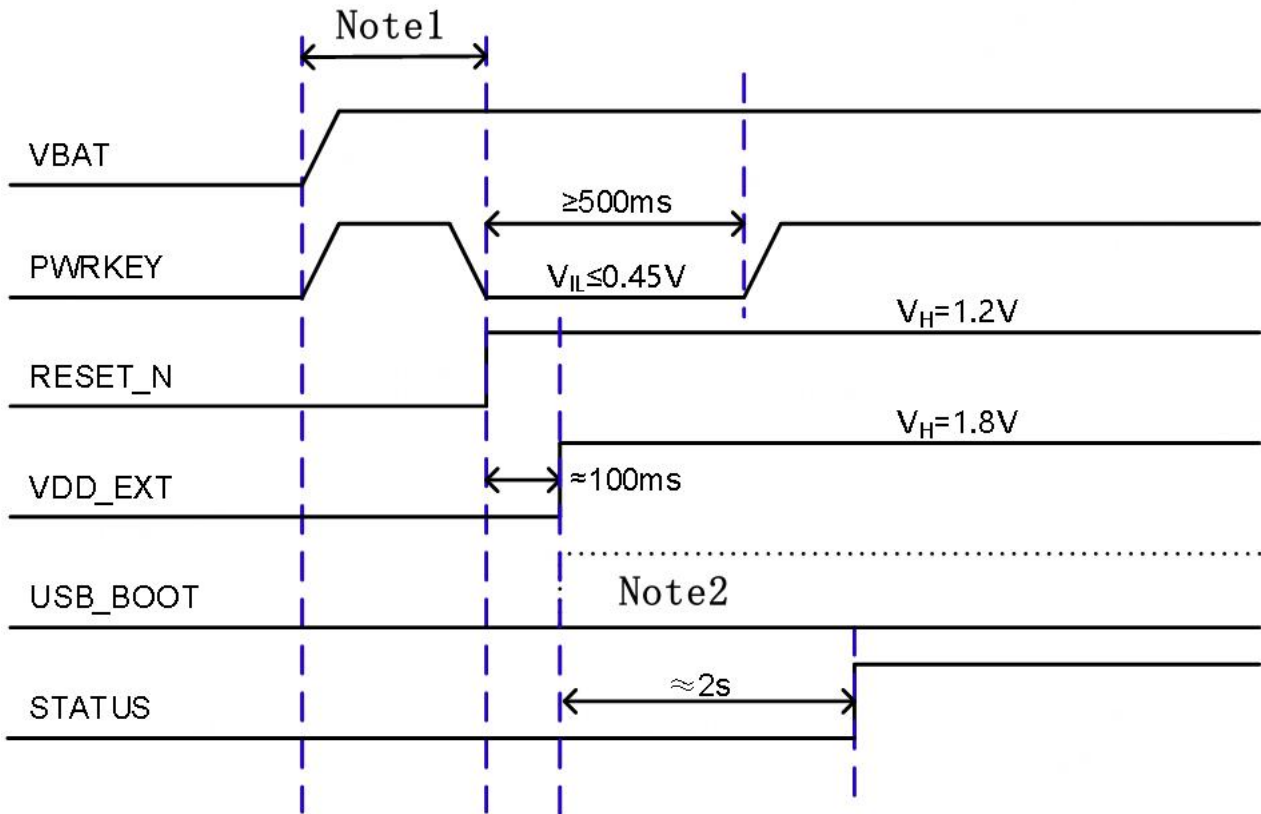


Figure 3-6 Power-On Sequence Diagram

Notes

◆ The initial state of VBAT when powered on needs to be less than 0.1V, and the time for VABT to rise from 0V to 2.5V should be less than 10ms; when pulling down Before the PWRKEY pin, ensure that the VBAT voltage is stable, and it is recommended that the time interval between powering on from VBAT to pulling down the PWRKEY pin is not less than 30ms.

◆ Pulling the USB_BOOT pin to VDD_EXT before booting the module will force the module into emergency download mode upon startup.

◆ If you need the device to power on automatically without the need for a power-off function, you can directly pull down the PWRKEY to GND. It is recommended to use a pull-down resistor of 4.7kΩ, or you can control the PWRKEY to be low level with GPIO before powering on the module and keep it low.

◆ The main serial port of the module sends the character "Lierda" to indicate that the boot-up is completed.

3.6 Shutdown

The module can be shut down in the following ways:

- ◆ Power off the module by controlling it through the PWRKEY pin.
- ◆ Send the command AT+QPOWD to power off.

3.6.1 PWRKEY shutdown

In the power-on state, pull down PWRKEY for 650ms and then release it, the module will execute the shutdown process, the timing is as follows:

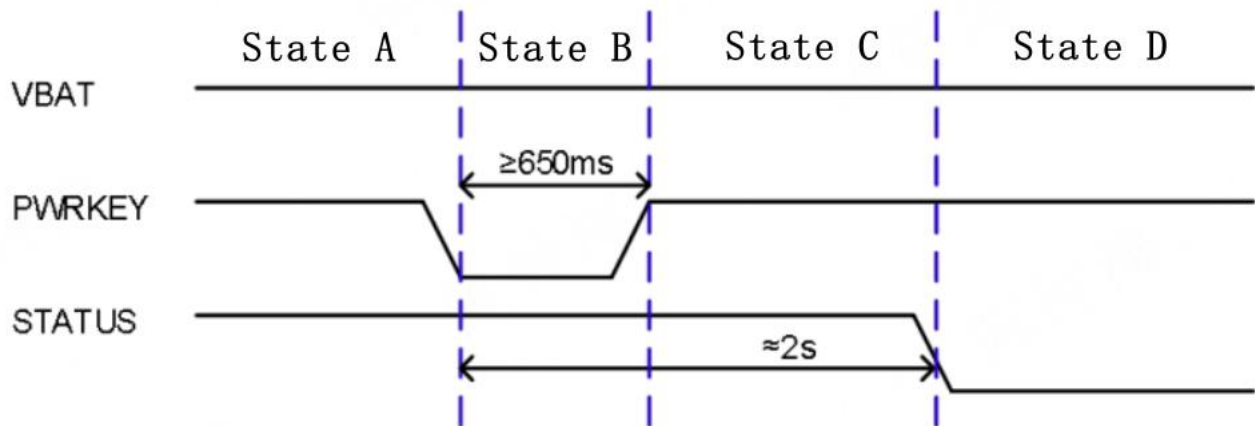


Figure 3-7 Power Key Shutdown Timing Diagram

PWRKEY shutdown timing description:

- ◆ Status A: Module operating status;
- ◆ Status B: Lower the status of PERKEY, for a minimum of 650ms.
- ◆ Status C: The system shuts down inside the module, from pulling down PWRKEY to shutdown completion takes about 2 seconds;
- ◆ Status D: The module has entered the shutdown state.

3.6.2 Power off using AT command

Executing AT+QPOWD can turn off the module, and this operation has the same timing and effect as turning off by pulling down PWRKEY.

Notes

◆ Do not cut off the power supply to the module immediately when the module is working normally, otherwise it may damage the Flash data inside the module.

It is recommended to turn off the module first using the PWRKEY or AT command, and then disconnect the power supply.

◆ When using the AT command to shut down, please ensure that after the shutdown command is executed, the PWRKEY remains at a high level, otherwise there may be a situation of the device turning on again after shutdown, which can damage the Flash if the power is disconnected at this time.

If the device is powered on by connecting PWRKEY to GND, it is recommended to use AT+QPOWD to perform the shutdown process before disconnecting the power supply.

3.7 Reset

3.7.1 Reset introduction

Pulling down RESET_N for at least 300 ms before releasing it can reset the module.

The description of the RESET_N pin is as follows:

Table 3-4 Reset Pin Description

Pin number	Pin Names	I/O	Description	DC characteristics	Note
15	RESET_N	DI	Module reset	VILmax=0.3V	There is weak pull-up internally, low level effective, leave floating if

					not used.
--	--	--	--	--	-----------

3.7.2 Reset circuit reference design

The RESET_N signal is sensitive to interference, it is recommended that the wiring of the module interface should be as short as possible, and grounding should be included. The typical reference hardware circuit applications are as follows, and you can choose between the two circuit diagrams based on actual needs:

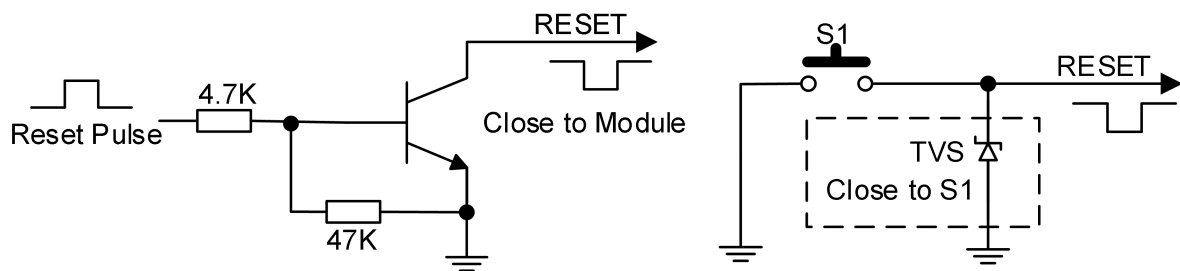


Figure 3-8 Reset Circuit Reference Design

3.7.3 Reset timing

The reset timing sequence of the module is as follows:

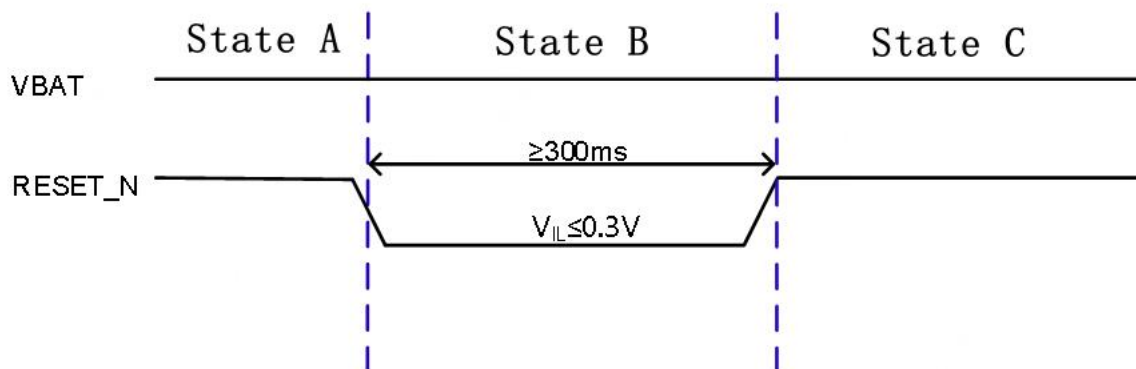


Figure 3-9 Reset Timing Diagram

Reset timing description:

- ◆ Status A: Module in normal working condition;
- ◆ State B: Lower the status of RESET_N for at least 300ms;
- ◆ Status C: The module is starting to reboot.

4 Application interface

NT26-KCN F series modules provide a wealth of peripheral functions, mainly including the following application interfaces:

- ◆ USB 2.0 interface
- ◆ UART interface
- ◆ USIM Interface
- ◆ I2C interface
- ◆ SPK interface

4.1 USB interface

NT26-KCN F series modules support USB2.0 interface, comply with USB2.0 specification, and support Full-Speed (12 Mbps) and High-Speed (480 Mbps) modes. The module only supports USB slave mode. This interface can be used for AT command sending, data transmission, software debugging, and firmware upgrading. The interface definition is as follows:

Table 4-1 USB Interface Pin Description

Pin number	Pin names	I/O	Description	DC characteristics	Note
59	USB_DP	DIO	USB differential data (+)	-	If not used, leave floating
60	USB_DM	DIO	USB differential data (-)	-	If not used, leave floating
61	USB_VBUS	DI	USB inserted wakes up the module.	VDD18AON	When in sleep mode, inserting a USB will wake up the module.

4.1.1 USB circuit reference design

The module USB interface can be connected to either a USB connector or the MCU's USB interface. The reference schematic for connecting to the USB connector is as follows:

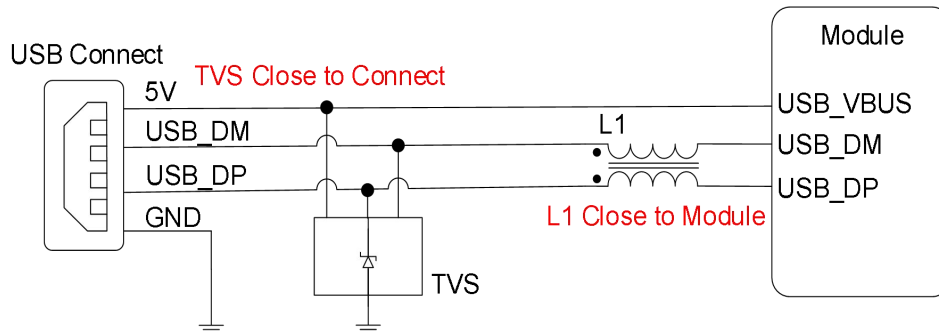


Figure 4-1 USB Interface Reference Design

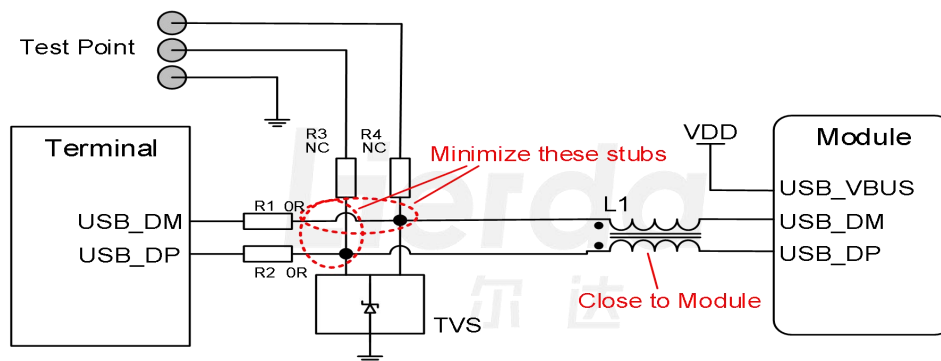


Figure 4-2 USB Connection MCU Reference Design

Notes

- ◆ To ensure the performance of USB, the USB_DP and USB_DM traces need to maintain a differential impedance of 90Ω, and the surrounding area is required. GND treatment. It is recommended to connect a common-mode inductor L1 between the MCU and the module to prevent USB signal from generating EMI interference. L1 needs to be placed close to the module.
- ◆ USB traces should be kept away from crystal oscillators, oscillators, magnetic devices, RF signals, and strong signal areas. It is recommended to use inner-layer differential traces and surround them with solid ground on all sides.

◆ If the USB interface of the module is connected to the MCU through a connector, TVS diode protection needs to be added at the interface. Special attention should be paid to the selection of ESD protection devices, and the parasitic capacitance should not exceed 2pF, and they should be placed as close to the USB interface as possible.

◆ Reserve resistors R1~R4, normally connect MCU by attaching R1~R2 and not attaching R3~R4; when debugging is needed, attach R3~R4 and not attach R1~R2 to switch to the test point. To meet the USB data line signal integrity requirements, keep the bifurcation lines connecting the test points as short as possible.

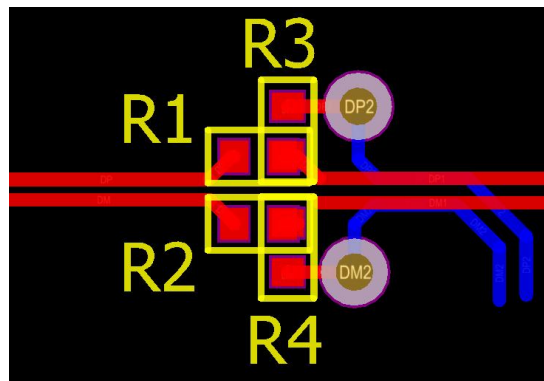


Figure 4-3 USB jumper routing diagram

4.1.2 USB data transfer

When the module is powered on or reset, if USB_BOOT is detected as a low level, the USB port will enumerate 2 serial ports for AT command sending and log output.

4.1.3 USB firmware download

When the module is powered on, restarted, or reset, if USB_BOOT is detected as a high level, the USB port will enumerate a dedicated download serial port. At this time, the download tool can upgrade the firmware through the USB interface. If no download tool is connected for more than 15 seconds, the module will re-enumerate 2 serial ports according to the default USB configuration.

The reference design for the USB_BOOT interface is as follows:

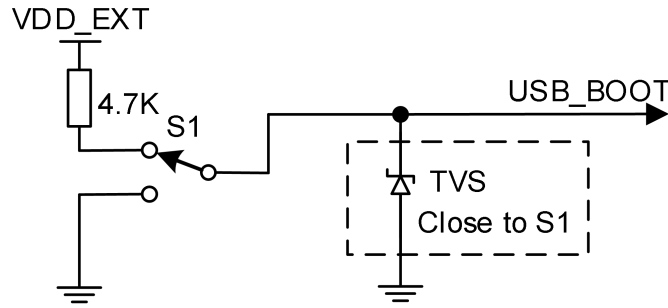


Figure 4-4 USB_BOOT Reference Design

4.2 UART communication

The module provides 3 general asynchronous transceivers: main serial port, debug serial port, and auxiliary serial port. The baud rate of the main serial port is configurable, and the debug serial port is only used for debugging and testing. The pin definitions for the serial ports are as follows:

Table 4-2 Serial Port Pin Definitions

Interface	Pin number	Pin names	Description	Note
Main serial port	17	MAIN_RXD	Main serial port receives data	Non-sleep state: VO_LDOIO power domain; Sleep status: Switch to internal AON power domain
	18	MAIN_TXD	Main serial port sends data	If not used, leave floating
	19	MAIN_DTR	Main serial port data terminal ready	In sleep mode, the wake-up pin can be used.
	20	MAIN_RI	Main serial port outputs ringing indication.	If not used, leave floating
	21	MAIN_DCD	Main serial port output	If not used, leave floating

			carrier detection	
	22	MAIN_CTS	Main serial port DTE clear to send	Connect CTS to DTE, leave floating if not used.
	23	MAIN_RTS	The main serial port DTE requests to send.	Connect RTS to DTE, leave floating if not used.
Debug serial port	38	DBG_RXD	Debugging serial port receiving data	If not used, leave floating
	39	DBG_TXD	Debug serial port to send data	If not used, leave floating
Auxiliary serial port	28	AUX_RXD	Assist in receiving data through the serial port.	Not supported at the moment.
	29	AUX_TXD	Assist in sending data through the serial port.	Not supported at the moment.

Notes

◆ The serial port interface belongs to the VO_LDOIO power domain, with a default voltage level of 1.8V;

◆ The VO_LDOIO power domain will lose power in sleep mode, and MAIN_RXD will switch to the internal 1.2V power supply.

- The RXD pin of the serial port has an internal pull-up and does not require an external pull-up resistor. It does not have reverse connection protection function.

◆ Pay attention to the consistency of the voltage levels when using the serial port, otherwise it may cause leakage current.

4.2.1 Serial Port Application

Main serial port characteristics:

◆ Used for AT command communication and data transmission, with a default baud

rate of 115200bps;

- ◆ Configurable as: 4800/9600/19200/38400/57600/115200/230400/460800bps;
- ◆ Used for firmware upgrade, the default baud rate during the upgrade is 921600bps;
- ◆ Support low power consumption wake-up module function.

Characteristics of serial port debugging:

- ◆ With the dedicated tools provided by the platform, you can view log information for software debugging, defaulting to 3000000bps.

Serial port connection diagram is as follows:

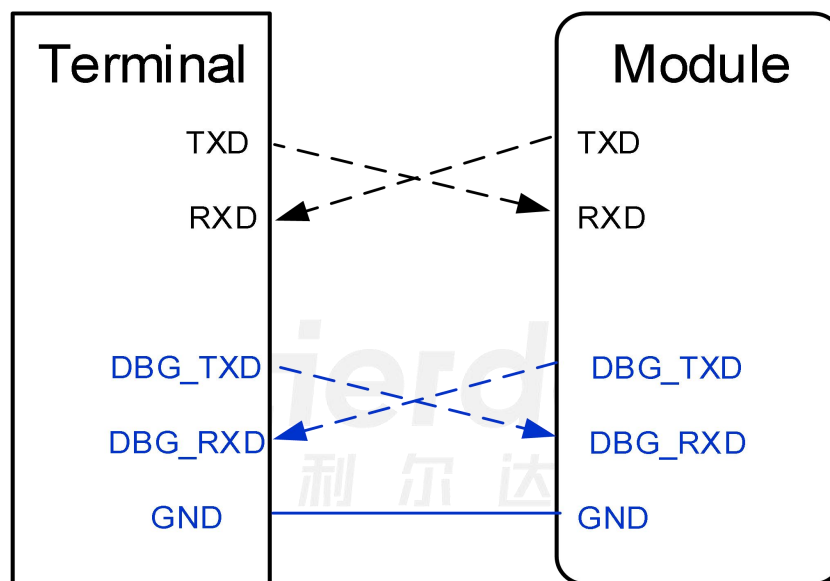


Figure 4-5 Serial Port Connection Diagram

4.2.2 Reference design for serial port circuit

The main considerations for a suitable serial port level conversion circuit include whether it meets the working speed of the serial port, scenarios with low power requirements, and whether the power consumption meets the requirements. It is recommended to refer to the following level matching circuit:

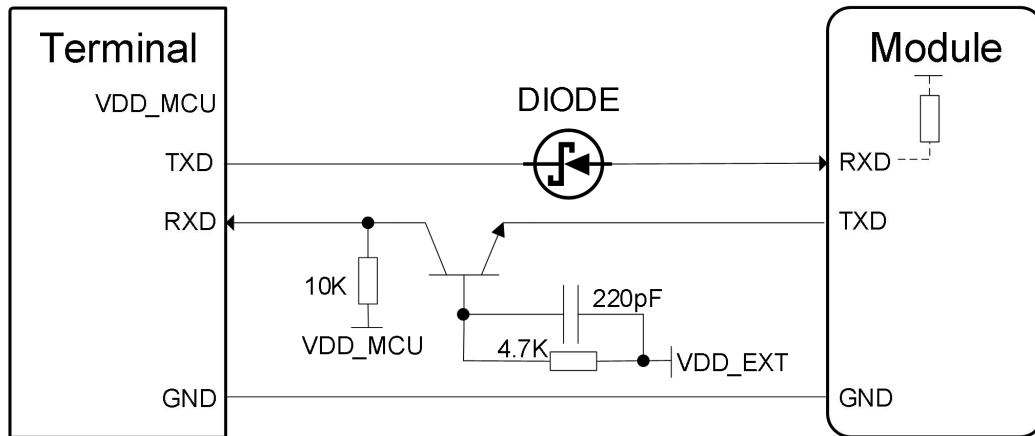


Figure 4-6 Transistor Level Conversion Reference Circuit

Recommend three transistors for reference:

Brand: CJ Model: S8050 J3Y Package: SOT-23

Recommended diode for reference (please note that the forward voltage of the diode should be $\leq 0.3V$):

Brand: LRC Model: LRB520S-30T1G Package: SOD-523

4.3 USIM card interface

The module includes 2 external USIM card interfaces, supports dual SIM single standby, complies with ETSI and IMT-2000 standards, and the main card supports 1.8V and 3.0V USIM cards. The pin description is as follows:

Table 4-3 Definition of External USIM Card Interface Pins

Pin number	Pin names	Description	DC characteristics
11	USIM_DATA	SIM card data cable	VO_LDOSIM
12	USIM_RST	SIM card reset line	VO_LDOSIM
13	USIM_CLK	SIM card clock line	VO_LDOSIM
14	USIM_VDD	SIM card power supply	Vnorm=1.8/3.0V
79	USIM_DET	SIM card status signal detection	SIM card slot hot plug detection interface

62	USIM2_CLK*	SIM2 (secondary card) clock line	VO_LDOIO
63	USIM2_RST*	SIM2 (sub card) reset line	VO_LDOIO
64	USIM2_DATA*	SIM2 (secondary card) data cable	VO_LDOIO
65	USIM2_VDD	SIM2 (secondary card) power supply,	Vnorm=1.8V

The module supports hot-swappable USIM cards through USIM_DET, with high level effective by default, and can be configured for high/low level detection. If the USIM card detection function is not required, USIM_DET can be left floating.

4.3.1 USIM card circuit reference design

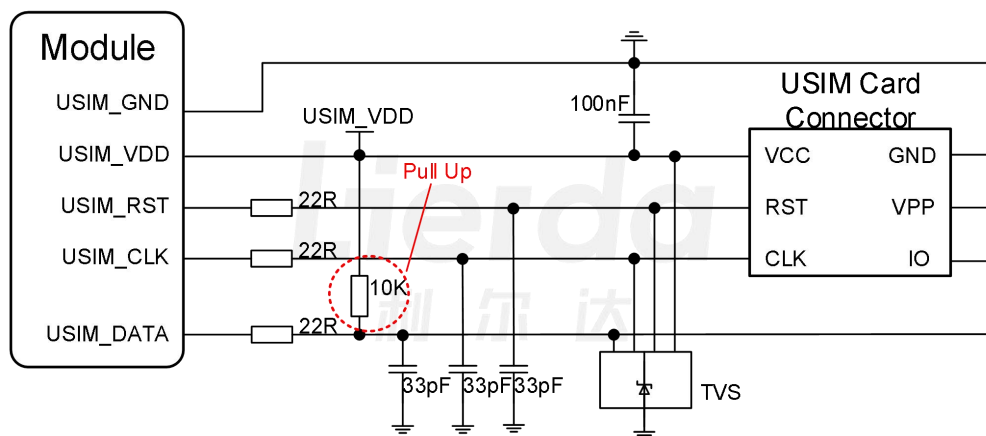


Figure 4-7 6PIN SIM card reference circuit

Notes

The hardware design circuit of USIM2 can also refer to the above design, specifically refer to the "Lierda NT26-KCN F series hardware reference design manual".

4.3.2 Notes on USIM card circuit design

To ensure the reliability and availability of the SIM card in applications, please follow the following USIM circuit design guidelines:

◆ When laying out, try to place the SIM card as close to the module as possible, and keep the signal line wiring length within 200mm if possible.

◆ SIM card signal line should be kept away from RF traces and VBAT power line;

◆ The GND wiring of the SIM card should be short and thick, ensuring that the wiring width is not less than 0.5mm;

◆ The decoupling capacitance of USIM_VDD should not exceed 1uF, and the capacitance should be placed close to the VCC of the external SIM card.

◆ To avoid signal interference between USIM_DATA and USIM_CLK, the wiring between them should not be too close. Ground shielding should be added between the two traces, and to prevent the negative impact of long traces, a pull-up resistor is usually added to USIM_DATA to USIM_VDD to enhance the driving capability. If the traces are too long, it is recommended to externally reserve a 10K resistor placed near the card slot. In addition, the USIM_RST signal also needs to be protected by grounding.

◆ To ensure good ESD protection performance, it is recommended to add a TVS diode to the external USIM card pins, and the parasitic capacitance of the selected TVS diode should not exceed 15pF; a 22 ohm resistor should be connected in series between the signal lines of the module and the SIM card to suppress stray EMI and enhance ESD protection. Additionally, a parallel 33pF capacitor should be used to filter out RF interference. The relevant resistors, capacitors, and TVS diode should be placed near the USIM card slot.

4.4 I2C interface

The module provides 1 I2C interface, the interface description is as follows:

Table 4-4 I2C Interface Pin Definitions

Pin number	Pin name	Description	DC characteristics	Note
57	I2C0_SDA	I2C data	VO_LDOIO	External pull-up required.
58	I2C0_SCL	I2C Clock	VO_LDOIO	External pull-up is needed.
66	I2C1_SDA	I2C data	VO_LDOIO	External pull-up required.
67	I2C1_SCL	I2C clock	VO_LDOIO	External pull-up required.

4.5 SPK interface

The module provides an analog audio interface that can drive low-power headphones.

Its pin definitions are as follows:

Table 4-5 SPK Pin Definitions

Pin number	Pin name	Description	DC Characteristics	Note
5	SPK_P	Audio differential output	VO_LDOIO	If not used, leave floating
6	SPK_N	Audio negative differential output	GND	If not used, leave floating

4.5.1 SPK reference application

High power speaker playback can be achieved through an external audio PA. The common reference circuit is as follows:

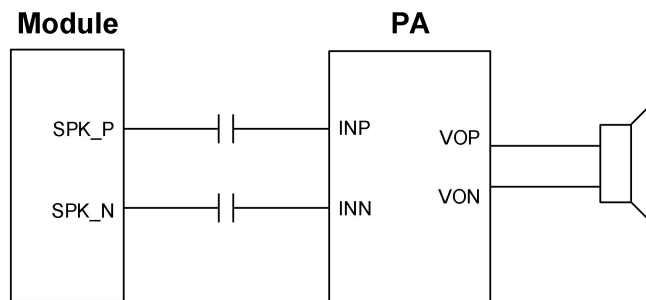


Figure 4-8 SPK Application Reference Circuit

In actual applications, the selected PA will determine the circuit design. For specific circuits, please refer to the "Lierda NT26-KCN F Series Hardware Reference Design Manual" document.

4.6 ADC interface

The module provides a 12-bit analog-to-digital conversion input interface (ADC interface). When wiring the ADC interface, in order to improve the voltage measurement accuracy of the interface, it is recommended to perform grounding treatment. The pin definitions are as follows:

Table 4-6 ADC Pin Definitions

Pin number	Pin name	Description	DC characteristics	Note
9	ADC0	General ADC interface	Internal direct connection Input voltage range: 0~1.05V	If not used, leave floating
96	ADC1	Universal ADC interface	Internal partial pressure Input voltage range: 0~3.3V Default 1.05V Internal partial	If not used, leave floating

			compression can be enabled through the software.	
--	--	--	--	--

Notes

If the acquisition voltage is greater than 3.3V, it is recommended to use a resistor divider circuit input for the ADC pin, and the recommended resistance value for the divider circuit is.

Between 100 kΩ and 1 MΩ, it is recommended to use resistors with an accuracy of 1% or higher, otherwise it will reduce the measurement accuracy of the ADC.

◆ When designing, reserve a 1nF capacitor at both ends of the ground potential divider resistor, by default, without soldering.

4.7 Status indicator signal

The module supports network status and operation status indicators, the interface is defined as follows:

Table 4-7 Definition of Status Indicator Pins

Pin number	Pin name	Description	DC characteristics	Note
16	NET_STATUS	Network status indicator	LDO_AONIO	If not used, leave floating
25	STATUS	Operating status indicator	LDO_AONIO/ VDD18AON	AGPIO function, power domain LDO_AONIO, no power loss during sleep. WAKEUP function, power domain VDD18AON, no power loss during sleep.

4.7.1 Network status indicator

The NET_STATUS pin is used to indicate the network status of the module, with the logic levels for different states as follows:

Table 4-8 Definition of Status Indicator Pins

Pin names	Level logic	Network status
NET_STATUS	Slow flash (200ms high/1800ms low)	Search network status
	Slow flash (1800ms high/200ms low)	Standby mode
	Quick Flash (125ms high/125ms low)	Data transmission mode

4.7.2 Running status indicator

The STATUS pin is used to indicate the operating status of the module. When the module is powered on normally, STATUS will output a high level, and it remains valid even in sleep mode.

4.7.3 Indicator light circuit reference design

NET_STATUS and STATUS pins can both be used to drive the LED indicator light, the reference design is as follows:

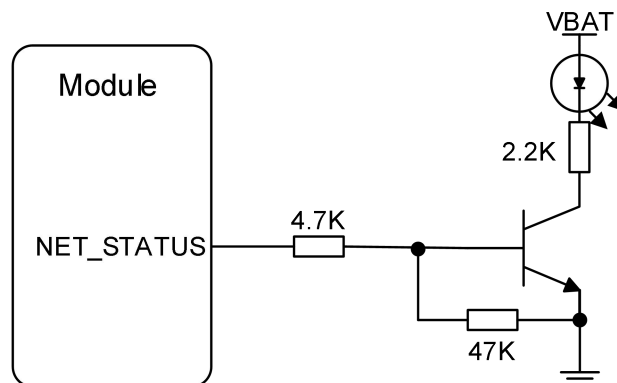


Figure 4-9 Indicator Light Circuit Reference Circuit

5 Radio Frequency Characteristics

5.1 Introduction to RF Function

NT26-KCN F series module provides one RF interface, with interface characteristic impedance of 50Ω. The interface description is as follows:

Table 5-1 RF Interface Pin Definitions

Pin number	Pin name	I/O	Description	DC characteristics	Note
35	ANT_MAIN	AIO	Main antenna interface	-	Used for air interface data transmission

5.2 Cellular network

5.2.1 Working frequency band

The operating frequency bands supported by the NT26-KCN F series module are as follows:

Table 5-2 Working Frequency Bands

Frequency band Band	Uplink frequency band Uplink(UL)band	Downlink frequency band Downlink(DL)band	Network standard Duplex Mode
1	1920MHz-1980MHz	2110MHz-2170MHz	FDD
3	1710MHz-1785MHz	1805MHz-1880MHz	FDD
5	824MHz-849MHz	869MHz-894MHz	FDD
8	880MHz-915MHz	925MHz-960MHz	FDD

5.2.2 Transmit power

The transmission power of the NT26-KCN F series module is as follows:

Table 5-3 Transmission Power

Band	Network Standard	Maximum value	Minimum value	Note
1	FDD	23dBm	<-40dBm	Compliant with Cat.1 protocol in 3GPP Rel-13 and Rel-14.
3	FDD	23dBm	<-40dBm	
5	FDD	23dBm	<-40dBm	
8	FDD	23dBm	<-40dBm	

5.2.3 Sensitivity of reception

The receiving sensitivity of the NT26-KCN F series module is as follows:

Table 5-4 Conduction reception sensitivity under single transmission (throughput ≥ 95%)

Band	Network standard	Receive sensitivity (10M)	3GPP standard (10M)
1	FDD	-97	-96.3dBm
3	FDD	-97	-93.3dBm
5	FDD	-98	-94.3dBm
8	FDD	-99	-93.3dBm

5.3 Antenna reference circuit design

When using the NT26-KCN F series module, a π -type matching circuit needs to be added between the RF antenna interface of the module and the antenna interface of the user's baseboard. The typical antenna matching circuit and initial parameters are shown in the following figure:

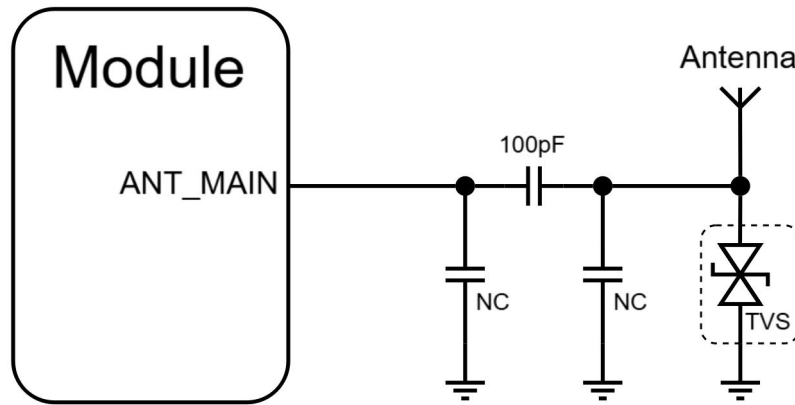


Figure 5-1 RF Antenna Reference Circuit

Notes

- ◆ Connect 100pF in series, and do not attach the parallel capacitor by default. Meanwhile, the recommended package for the device is 0201 or 0402.
- ◆ If an external antenna is used, or if the antenna can be touched by the user, it is recommended to reserve a TVS diode for enhanced electrostatic protection. Because the parasitic capacitance of the TVS may affect the antenna performance, it is recommended to re-adjust the antenna after adding the TVS diode.

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5.4 RF signal cable routing guidance

The wiring between the module antenna interface and the user antenna must meet the 50-ohm characteristic impedance requirements. At the same time, the RF wiring should be kept as short as possible to ensure minimal insertion loss of the RF wiring.

In general, the impedance of RF signal lines is determined by the material's dielectric constant ϵ_r , trace width W , spacing to GND D , and the thickness of the reference GND plane H .

In the field of IoT applications, the design of PCB characteristic impedance is usually achieved using coplanar waveguide technology, which helps to achieve better shielding for RF signal lines and enables higher integration for small footprint designs.

Common co-planar waveguide designs for PCBs include the following:

Table 5-5 Description of Common PCB Coplanar Waveguide Structures

Serial number	Pin name	Note
1		2-layer board Reference ground is at layer L2.
2		4-layer board Reference ground is at level L3. L2 layer excavation
3		4-layer board Reference ground is at layer L4. Excavation at levels L2 and L3

During the layout process, a 50-ohm impedance can be simulated and calculated using the Polar Si9000 tool. Based on actual conditions, appropriate line width, spacing, and stackup can be selected to achieve the best design effect.

Using a 2-layer, 1.6mm thick PCB as an example to explain co-planar waveguide calculations:

◆ Choose the appropriate PCB substrate material, once the substrate is selected, the dielectric constant ϵ_r is basically determined (in this case, it is 4.6);

◆ To choose the appropriate line width, a good method is to select a line width that matches the size of the pad left by the component (note that it should not be too thin, choose 0.65mm in this example);

◆ Based on the selected board thickness, line width, dielectric constant, etc., the spacing to ground can be roughly calculated. If the line spacing is not appropriate, the line width can be adjusted to modify the line spacing (calculated to be 0.14mm, which is basically appropriate).



Reference for calculating the impedance of 50 ohms in Figure 5-2.

During the actual layout process, the following suggestions are provided for reference:

In a π -type circuit, three matching reserved components are placed close to the antenna, and bypass components are recommended to be placed on both sides of the RF line.

◆ Irregular vias are often placed on both sides of the RF traces, and there must be a complete GND plane below the entire RF trace space.

◆ There should be no other traces below the RF line to avoid affecting the RF performance or other circuits.

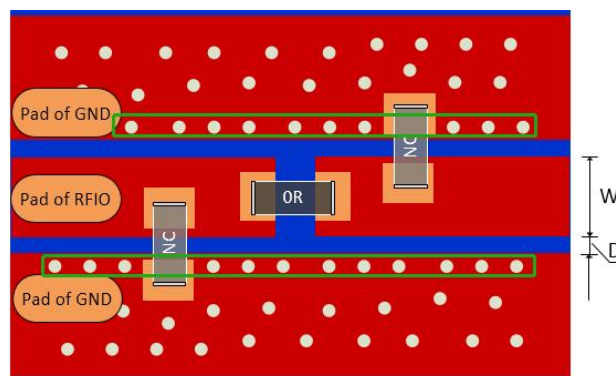


Figure 5-3 RF routing LAYOUT design schematic

The reasonableness of RF routing can be directly reflected in the module's conducted tests. In order to ensure the full performance of the product, antenna design cooperation is also required. In order to better meet the requirements of antenna design, the following requirements are hoped to be achieved in PCB design. Guidance will be provided for whole machine PCBs of different layers:

- When the product PCB is designed with 2 layers, it is best for both the TOP and BOTTOM LAYER directly below the module to be GND layers. The traces that the module needs to lead out should avoid running directly below the module and should be routed out from the outer side of the module.

- ◆ When the product PCB is designed with 4 layers, it is recommended that the traces that need to be led out by the module should be routed on the third or fourth layer, leaving the first and second layers for the module to have a complete GND reference layer.

5.5 Antenna design requirements

The RF performance of the module is also affected by the antenna. The selection of the antenna needs to meet the following requirements:

- ◆ Choose an antenna that matches the operating frequency band of the module.
- ◆ The impedance of the antenna is required to be 50 ohms to reduce the loss at the connection between the RF cable and the antenna.
- ◆ The lower the insertion loss within the working frequency band, the better, such as $VSWR \leq 2$, return loss $\geq 10\text{dB}$, etc.;

Commonly used antennas suitable for CAT1 scenarios are as shown in the following figure:

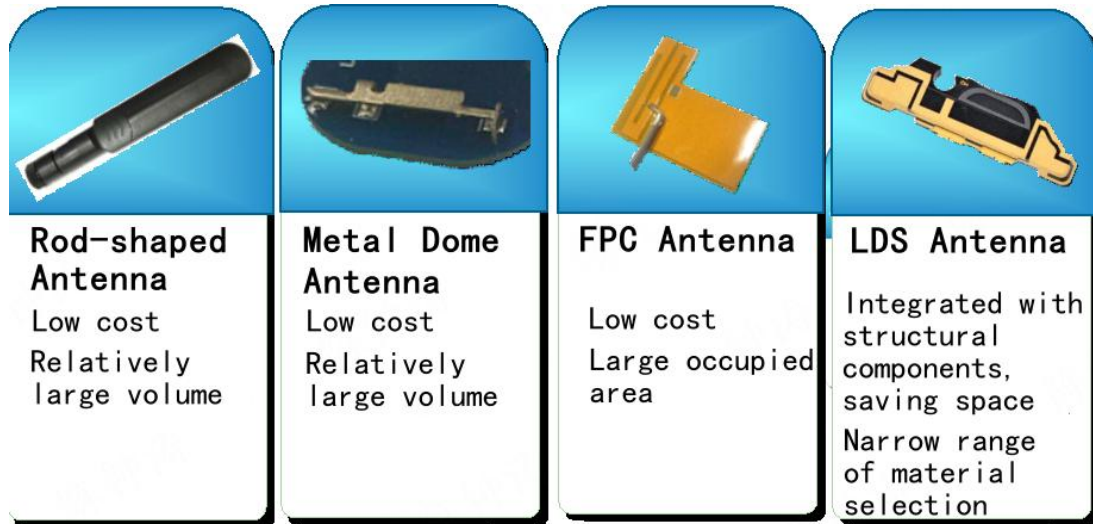


Figure 5-4 Commonly Used Antenna Types for CAT1

5.6 Recommended RF connectors

In some applications where customized antennas are needed, such as in environments where devices are shielded by metal casings, leaving IPX or SMA connectors on the motherboard can help connect the antenna outside the device through an RF cable to achieve better performance.

Table 5-6 Common RF Connectors

Serial number	Product name	Image	Note
1	IPX socket		Connect the RF cable.
2	SMA constellation		Can connect to RF cable or antenna.

6 Electrical performance and reliability

6.1 Absolute maximum rating

The description of the absolute maximum rated value of the module interface is as follows:

Table 6-1 Absolute Maximum Ratings

Parameters	Description	Minimum value	Maximum value	Unit	Note
VBAT	Module power supply voltage	-0.3	5.0	V	
VI _{AIO}	ADC input voltage	-0.3	3.6	V	
VI _{GPIO}	GPIO input voltage	-0.3	3.6	V	
Others	Other pin input voltage	-0.3	3.6	V	

Precautions

When the operating conditions exceed the absolute maximum ratings, it may cause permanent damage to the module.

6.2 Rated power value

The electrical characteristics of the module power interface are described as follows:

Table 6-2 Power Interface Electrical Characteristics

Parameters	Description	Minimum value	Typical value	Maximum value	Unit
VBAT	Module normal supply voltage(1)	3.3	3.8	4.5	V
VBAT	Module expansion power supply voltage (2)	3.1	3.8	4.5	V
USIM_VDD	USIM card power supply output	-	1.8/3.0	-	V

USIM2_VDD	USIM2 card power supply output	-	1.8	-	V
VDD_EXT	Reference power output (3)	-	1.8	-	V

Notes

◆ When the module operates within this voltage range, the module's relevant performance meets the requirements of the 3GPP standard.

◆ When the module operates within this voltage range, it can still maintain normal operation without irreparable failures; only individual indicators, such as output power and other parameter values, may exceed the range of the 3GPP standard. When the voltage returns to the normal operating range, all indicators of the module still comply with the 3GPP standard.

◆ Sleep Mode: VDD_EXT power off, driving capability 120mA; VDD_EXT not power off, driving capability 3mA.

6.3 Power consumption

Power consumption of the module is as follows:

Table 6-3 Module Power Consumption

Module description	Test conditions	Typical value	Unit
Shutdown mode	Module shutdown	0.5	uA
Sleep mode	AT+CFUN=0 (USB disconnected)	2.5	uA
	LTE-FDD @ PF = 32 (USB disconnected)	0.76	mA
	LTE-FDD @ PF = 64 (USB disconnected)	0.32	mA
	LTE-FDD @ PF = 128 (USB disconnected)	0.25	mA
	LTE-FDD @ PF = 256 (USB disconnected)	0.18	mA
Idle mode	LTE-FDD @ PF = 64 (USB disconnected)	4	mA
LTE data transmission	LTE-FDD B1	580	mA
	LTE-FDD B3	610	mA

	LTE-FDD B5	550	mA
	LTE-FDD B8	550	mA

6.4 Digital logic level characteristics

The logic level definitions of the module's digital IO are as follows:

Table 6-4 Explanation of Digital IO Logic Levels

Type	Parameters	Description	Minimum value	Maximum value	Unit
Input	VIL	Low level input	-	$0.2 \cdot V_{DD_EXT}$	V
	VIH	Input high level	$0.7 \cdot V_{DD_EXT}$	-	V
Output	VOL	Low level output	-	$0.15 \cdot V_{DD_EXT}$	V
	VOH	Output high level	$0.8 \cdot V_{DD_EXT}$	-	V

Notes

The GPIO reference level of the module follows V_{DD_EXT} , and V_{DD_EXT} will lose power during sleep, so will the corresponding GPIO.

Power off.

The logic voltage levels of the USIM card interface module are defined in the following table:

Table 6-5 USIM Logic Level Description

Type	Parameters	Description	Minimum value	Maximum value	Unit
Input	VIL	Low level input	-	$0.2 \cdot USIM_V_{DD}$	V
	VIH	Input high level	$0.7 \cdot USIM_V_{DD}$	-	V
Output	VOL	Low level output	-	$0.15 \cdot USIM_V_{DD}$	V
	VOH	Output high level	$0.8 \cdot USIM_V_{DD}$	-	V

Notes

The voltage level of USIM_VDD will be determined according to the detected category of the USIM card, supporting 1.8/3.0V USIM cards.

6.5 Static electricity protection

In life and production, static electricity is everywhere. Static electricity generated by human body static electricity, object friction, etc., may be transferred to the module through various ways, and may cause damage to the module. Therefore, it is necessary to pay great attention to static electricity protection and take necessary static electricity protection measures. For example:

- ◆ Wear anti-static gloves during the R&D, production, assembly, and testing processes;
- ◆ When designing products, add anti-static protection devices at circuit interfaces and other points vulnerable to electrostatic discharge.

The electrostatic discharge performance of the module is as follows:

Table 6-6 Electrostatic Protection Characteristics

Test interface	Discharge phenomenon	Note
VBAT	±8KV	Test standard: IEC61000-4-2 Temperature: 25°C Humidity: 45%
GND	±8KV	
Antenna interface	±8KV	

6.6 Working and storage temperature

Table 6-7 Operating Temperature Range

Parameters	Minimum value	Typical value	Maximum value	Unit	Note
Normal operating temperature ⁽¹⁾	-35	+25	+75	°C	
Expand operating temperature ⁽²⁾	-40	+25	+85	°C	
Storage environment temperature ⁽³⁾	-40	+25	+90	°C	

Notes

(1)When the module operates within this temperature range, its relevant performance meets the 3GPP standard requirements.

(2)When the module operates within this temperature range, it can still maintain normal operation without irreparable failures; only individual indicators, such as output power and other parameters, may exceed the range specified in the 3GPP standard. When the temperature returns to the normal operating range, all indicators of the module still comply with the 3GPP standard.

(3)This storage temperature range does not include packaging materials, and it is important to note the maximum temperature tolerance of the reel tape packaging.

6.7 Notes

During the production and use of modules, there are some points to note:

◆ When spraying the module, please try to avoid the spraying material from flowing into the interior of the module.

- When cleaning the module, do not use ultrasonic cleaning on the module, as it may cause damage to the internal crystals of the module.

◆ When manufacturing and using modules, avoid applying them in environments or enclosures containing any amount of mercury or mercury vapor, as this may lead to product failure or malfunction risks.



7 Mechanical dimensions

7.1 Mechanical dimensions

The mechanical dimensional drawing of the module is as follows:

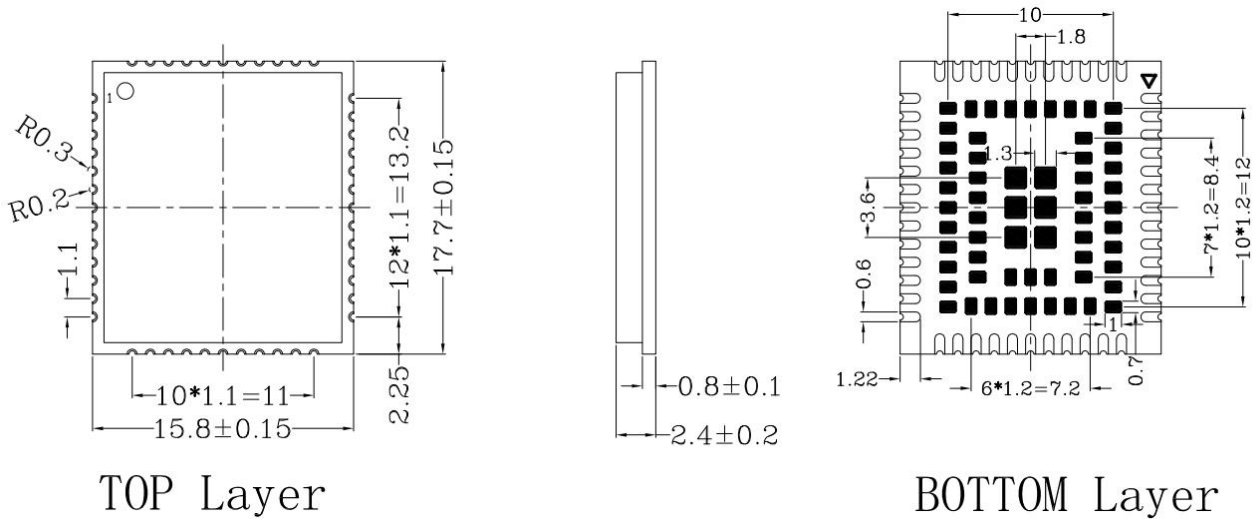


Figure 7-1 Module Mechanical Dimension Diagram (unit: mm)

7.2 Module top view / bottom view

The physical effect diagram of the module is as follows, please refer to the actual module for details:



Figure 7-2 Module Top/Bottom View

7.3 Recommend encapsulation

The module recommends solder pads as shown in the following figure, and users can make minor adjustments according to their own production processes.

◆ The pins around the module are designed with rounded corners inside. When designing the bottom board for soldering, please consider using rounded corners for transition. The rectangular solder pads on the bottom of the module can be designed using the module pin dimensions. Please refer to the design diagram of a single solder pad in the figure below.

◆ To facilitate the opening of the stepped steel mesh, it is recommended not to layout other components within 2.0mm range outside the solder pad of the module. Users can refer to the requirements of their own steel mesh manufacturers to determine this distance.

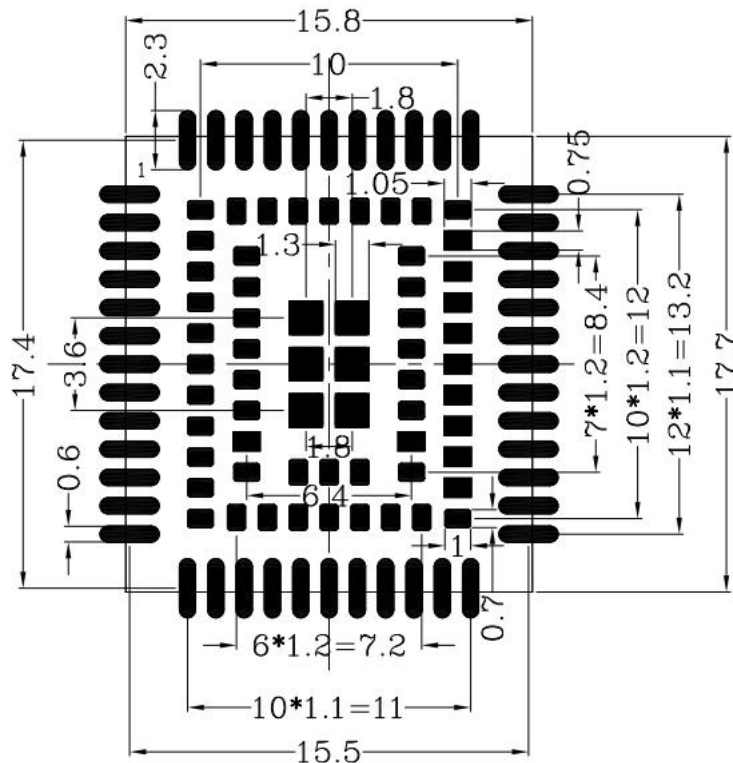


Figure 7-3 Module Recommended Solder Pads (unit: mm)

8 Production and packaging information

This chapter describes information such as the SMT process, packaging method, storage conditions of the module, which can help users better store and use this module.

8.1 Manufacturing welding

8.1.1 Furnace passing method

When the module is being surface mounted, the following points need to be noted during reflow soldering:

If the bottom of the module uses double-sided SMT, it is recommended to place the module in the second SMT process.

◆ It is best for the customer's PCB to go through the reflow oven on the mesh belt for the first SMT placement, and for the second SMT placement, it should also be passed through the reflow oven on the mesh belt as much as possible.

◆ If for special reasons it is not possible to put it on the net belt for reflow soldering, consider using fixtures for reflow soldering on the track or placing a flat high-temperature resistant straight template under the PCBA for reflow soldering to prevent PCB deformation during reflow soldering, which may lead to virtual soldering of the module.

8.1.2 Soldering Operation Guidance

The temperature curve of PCBA reflow soldering furnace is related to the use of solder paste and needs to be adjusted according to the actual solder paste. The data is only suitable for lead-free operation, refer to Figure 8-2 for lead-free reflow soldering operation guidelines.

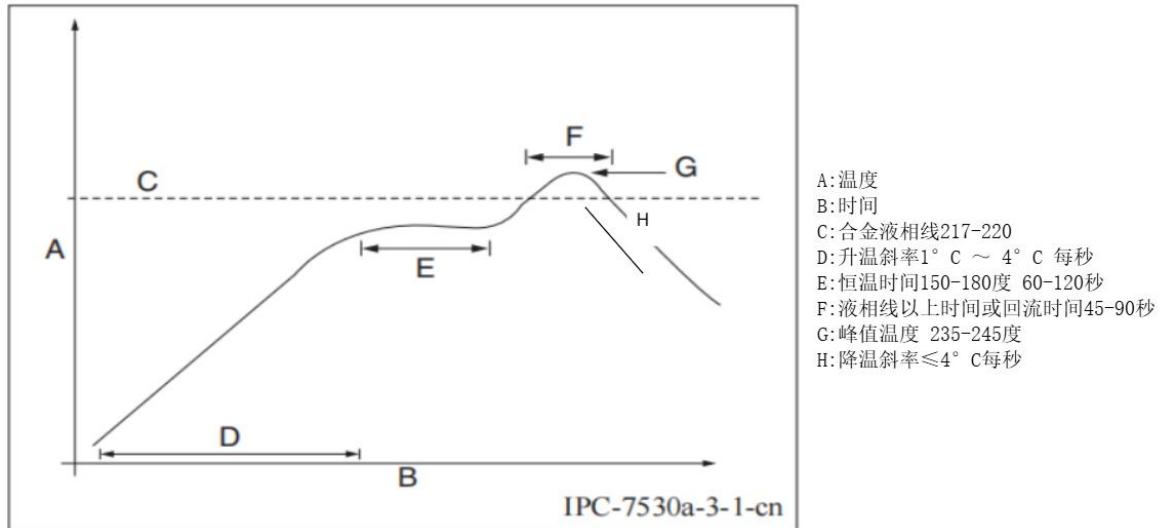


Figure 8-1 Lead-Free Reflow Soldering Operation Guide

8.1.3 Production process

During the production welding or any process that may directly contact the module, no organic solvents (such as alcohol, isopropanol, acetone, trichloroethylene, etc.) should be used to wipe the module shield; otherwise, it may cause the shield to rust.

If you need to spray or glue the module, please make sure that the spray or glue materials used will not chemically react with the module shield or PCB, and ensure that the spray or glue materials will not flow into the interior of the module.

8.1.4 Repair

If the module has poor soldering, short circuit or other defects that require repair, please follow the parameters below:

- ◆ Lead-free process: Soldering iron temperature $380 \pm 10^{\circ}\text{C}$, soldering iron contact time $\leq 5\text{S}$.
- ◆ Lead process: Soldering iron temperature $350 \pm 10^{\circ}\text{C}$, soldering iron contact time $\leq 5\text{S}$.

It is not recommended to use a hot air gun to blow on the module in order to avoid

affecting the module's performance.

8.2 Packaging specifications

The factory packaging of this module uses a rubber wheel carrying method, the reference dimensions of the rubber wheel are as follows:

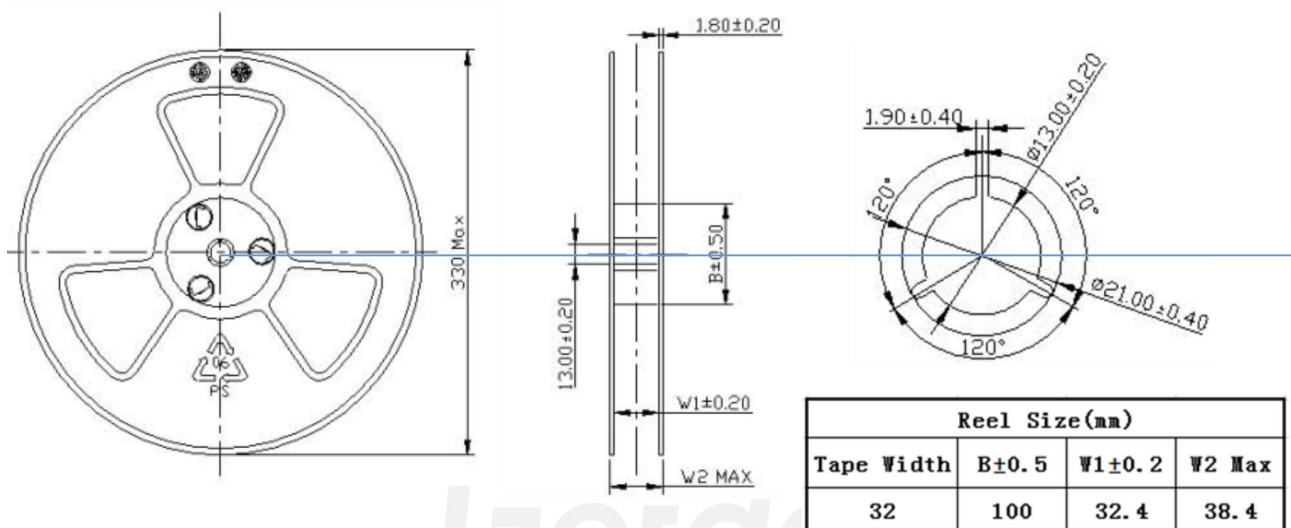


Figure 8-2 Carrying Belt Pulley Dimensional Drawing

The loading direction is as follows: (Refer to the diagram, the actual content of the label shall prevail, pay attention to the position of module PIN1)

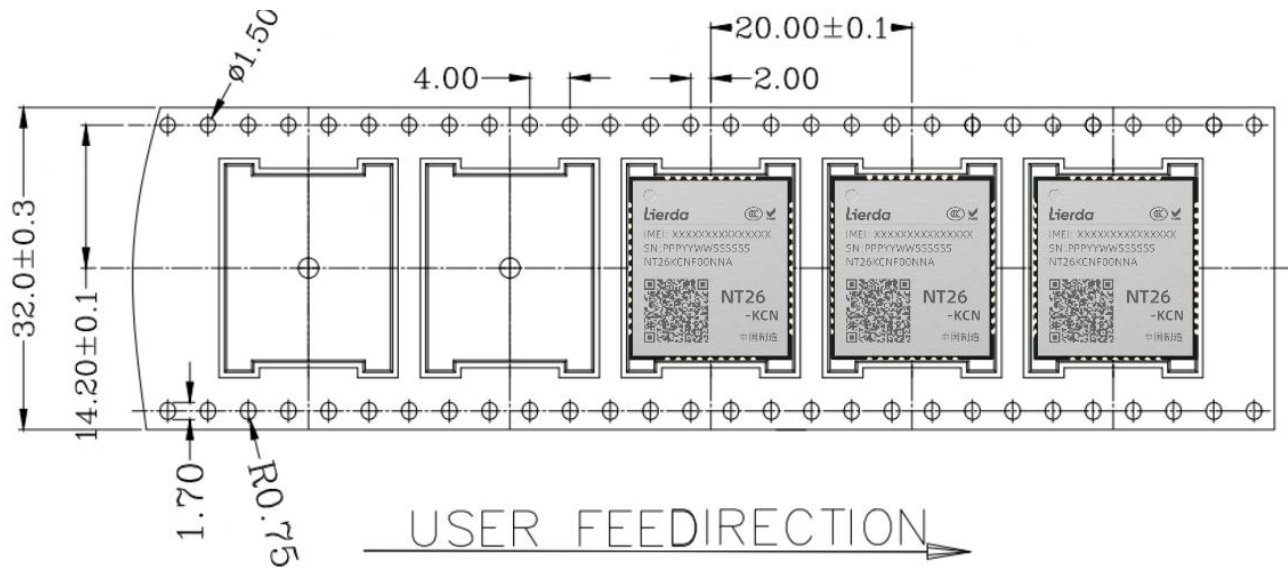


Figure 8-3 Loading Direction Indicator for Conveyor

8.3 Storage conditions

The module is shipped in the form of a vacuum reel sealed bag, with a humidity sensitivity level of MSL 3. Storage conditions are as follows:

- Temperature below 40°C, humidity below 90% (RH), can ensure 12 months of solderability in well-sealed packaging.

- ◆ After opening the package, ensure that surface mount assembly is carried out within 168 hours under the condition of temperature less than 30°C and relative humidity less than 60% (RH).

- ◆ If baking is required to meet the above conditions, the tape and reel components should be baked at 60°C±5°C, humidity ≤60%RH for 48 hours; if accelerated baking is needed, the modules need to be taken out from the tape and reel for individual baking (ESD protection should be observed during the removal process), baked at 125°C±5°C, humidity ≤60%RH for 8 hours, with a total baking time of less than 96 hours.

Please refer to the IPC/JEDEC J-STD-033 standard for more detailed guidance.

9 Relevant documents and terminology abbreviations

9.1 Related documents

Reference documents are as follows:

Table 9-1 Related Documents

Serial number	Document Name	Comments
[1]	Lierda NT26-KCN F Series Hardware Reference Design Manual	

9.2 Term abbreviation

The abbreviations and their corresponding explanations mentioned in this document are listed in the table below:

Table 9-2 Term Abbreviations

Abbreviation	Full English name	Chinese full name
3GPP	3rd Generation Partnership Project	The third generation Partner Program
ADC	Analog-to-Digital Converter	Analog-to-Digital Conversion
ANT	Antenna	Antenna
ASM	Antenna Switch Module	Antenna switch module
DAC	Digital -to- Analog Converter	Analog-to-digital conversion
DBG	Debug	Debugging
DC-DC	Direct Current - Direct Current	DC converter
DCXO	Digitally Controlled Crystal Oscillator	Numerically Controlled Crystal Oscillator
DRX	Discontinuous Reception	Non-continuous reception

DTE	Data Terminal Equipment	Data terminal equipment
ECL	Equivalent Class Level	Network coverage level
ESD	Electro-Static discharge	Electrostatic discharge
EOS	Electrical Overtress	Electrical super stress (surge)
ESR	Equivalent Series Resistance	Equivalent series resistance
EVK	Evaluation Kit	Assessment Toolkit
H-FDD	Half Frequency Division Duplexing	Frequency Division Half Duplex
FOTA	Firmware Over-The-Air	Remote firmware upgrade
GPIO	General-purpose input/output	Universal input and output
I/O	Input/Output	Input Output Interface
I _{max}	Maximum Load Current	Maximum current
I _{norm}	Normal Current	Normal (typical) current
bps	Bits Per Second	Units of speed
LCC	Leadless Chip Carriers	Leadless chip carrier package
LCM	LCD Module	LCD display module
LDO	Low Dropout Regulator	Low dropout linear regulator
LGA	Land Grid Array	Grid array packaging
LwM2M	Lightweight Machine-To-Machine	Lightweight M2M protocol
MCU	Mirco Controller Unit	Microcontrol unit
MSL	Moisture Sentivity levels	Wet sensitivity level
PCB	Printed Circuit Board	Printed Circuit Board
PCBA	Printed Circuit Board Assembly	Printed circuit board components
PMU	Power Management Unit	Power Management Unit
PSM	Power Saving Mode	Energy-saving mode

RF	Radio Frequency	Radio frequency
RoHS	Restriction of Hazardous Substances	Limitation of harmful substances
RX	Receive	Receive
TAU	Tracking Area Update	Track area update
TCP/IP	Transmission Control Protocol/Internet Protocol	Transmission Control Protocol/Internet Protocol
TVS	Transient Voltage Suppressor	Transient Voltage Suppression Diode
TX	Transmit	Send
UART	Universal Asynchronous Receiver & Transmitter	General asynchronous receiver and transmitter
UDP/IP	User Datagram Protocol - Internet Protocol	User Datagram Protocol
URC	Unsolicited Result Code	Non-request result code
(U)SIM	(Universal) Subscriber Identification Module	Universal user identity recognition module
VSWR	Voltage Standing Wave Ratio	Voltage Standing Wave Ratio
Vmax	Maximum Voltage Value	Maximum voltage
Vnorm	Normal Voltage Value	Normal (typical) voltage
Vmin	Minimum Voltage Value	Minimum voltage
VIHmax	Maximum Input High Level Voltage Value	Maximum input high level
VIHmin	Minimum Input High Level Voltage Value	Minimum input high level
VILmax	Maximum Input Low Level Voltage Value	Maximum input low level
VILmin	Minimum Input Low Level Voltage Value	Minimum input low level
VImax	Absolute Maximum Input Voltage Value	Maximum input level
VImin	Absolute Minimum Input Voltage Value	Minimum input level
VOHmax	Maximum Output High Level Voltage Value	Maximum output high level
VOHmin	Minimum Output High Level Voltage Value	Minimum output high level

VOLmax	Maximum Output Low Level Voltage Value	Maximum output low level
VOLmin	Minimum Output Low Level Voltage Value	Minimum output low level